Step by Step from Environmental Samples to Preservation Vials

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(Lab plain-PDF edition, rebuilt 2026-10-20)

Source of record: HuangLab (Yin-Tse Huang lab, NCKU). Originally authored on protocols.io (dx.doi.org/10.17504/protocols.io.n2bvj8mk5gk5/v1), created 2023-04-05, version 0. This PDF is the lab's self-hosted copy and supersedes the protocols.io export.

Abstract

This protocol shows how to generate a MICROBES ISOLATION FROM ENVIRONMENTAL SAMPLES into LAB COLLECTION. Please follow the instructions and do not modify the protocols by yourself. If any changes are needed please inform the lab manager or admin.

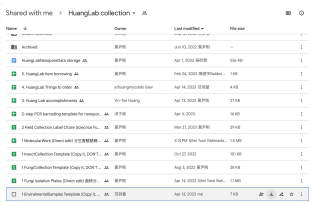
Procedure

Isolate Primary Culture from Environmental Samples

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1. Once you received your environmental samples.

Please catalog them in your Environmental Samples spreadsheet.



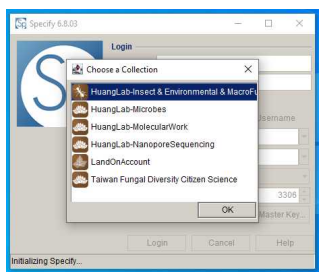
Make a copy of the template for yourself. The template "1 EnvironmentalSamples Template" marked with blue box is available in HuangLab Collection.

Upload your datasheet to Specify database

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3. ****The account setup is skipped****

Login to the Specify database. Choose the **first collection (HuangLab-Insect & Environmental & MacroFungi)**.



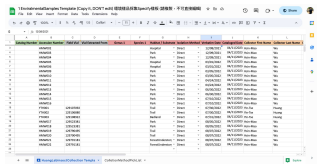
Isolate Primary Culture from Environmental Samples

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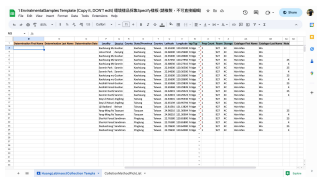
2. Fill in the columns of the spreadsheet step by step (As shown in the following steps 2.1-2.15)

*In the template, some template data are there for your reference, please **delete them in your copy** and fill in only your data.

- Remember to scroll to the right and fill in as much as information you can.

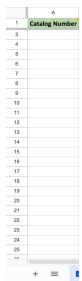


Column in the left

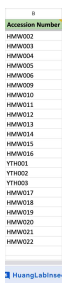


Scrolling to the right and fill them in

2.1. Leave the Catalog Number (Column A) blank. The Catalog Number will be generated automatically from Specify. Copy and paste it directly after the upload.



2.2. Accession Number (Column B) is for your own reference, you can name your sample as you want, there is no specific format for this.



2.3. Field Vial (Column C) is the iNaturalist code. You only need to fill this column when you got the samples from citizen scientists.

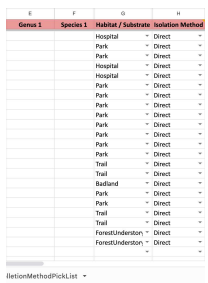


2.4. Leave the Vial Extracted From (Column D) blank.



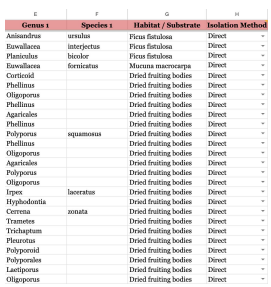
2.5. For Column E, F, G, G:

If your samples are **SOIL SAMPLES**, please follow this example.



This is the example for Soil Samples. Genus 1 & Species 1: Leave it blank. Habitat/Substrate: What is the environment you collected your samples. Isolation: How you isolate your samples (the default is Direct).

If your samples are **ENVIRONMENTAL SAMPLES**, please follow this example.



This is the example for Environmental Samples (e.g., ambrosia & bark beetles, fruiting bodies, etc). Genus 1 & Species 1: Fill in the scientific name of your samples (unable to identify just leave it blank). Habitat/Substrate: What is the environment you collected your samples (e.g., the host plant for the ambrosia & bark beetles). Isolation: How you isolate your samples (the default is Direct).

2.6. Fill in the Verbatim Date (Column I) with the date **you COLLECTED the samples**.

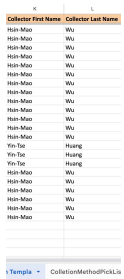
Fill in the Cataloged Date (Column J) with the date **you UPLOADED the data to Specify**.



Verbatim Date	Cataloged Date
12/08/2021	04/11/2023
12/09/2021	04/11/2023
12/09/2021	04/11/2023
03/01/2022	04/11/2023
03/01/2022	04/11/2023
03/05/2022	04/11/2023
03/05/2022	04/11/2023
04/24/2022	04/11/2023
04/24/2022	04/11/2023
06/30/2022	04/11/2023
06/30/2022	04/11/2023
07/10/2022	04/11/2023
07/10/2022	04/11/2023
07/30/2022	04/11/2023
07/30/2022	04/11/2023
07/31/2022	04/11/2023
08/05/2022	04/11/2023
08/05/2022	04/11/2023
08/07/2022	04/11/2023
08/07/2022	04/11/2023
08/07/2022	04/11/2023
08/07/2022	04/11/2023

2.7. Collector First Name (Column K) and Collector Last Name (Column L)

Fill in the name of the collector.



Collector First Name	Collector Last Name
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Yin-Tse	Huang
Yin-Tse	Huang
Yin-Tse	Huang
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu
Hsin-Mao	Wu

t Templa | CollectionMethodPickLis

2.8. Fill in these columns when the sample is identified with a scientific name, remember to fill Column E and F (Genus and Species names) as well. If not, leave it blank.

Fill in the date you did the determination (identification).



Determination First Name	Determination Last Name	Determination Date

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2.9. Fill in these columns with the geographical information of your samples.

Leave it blank, when you don't have the information or the materials is outsourced lacking this info.



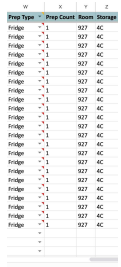
Locality	District	County	State/Province	Country	Latitude	Longitude
Kaohsiung Mu	Gushan		Kaohsiung	Taiwan	22.65500	120.29100
Lotus Pond	Zuoying		Kaohsiung	Taiwan	22.68300	120.29600
Kaohsiung Mu	Gushan		Kaohsiung	Taiwan	22.65600	120.28600
Kaohsiung Mu	Sanmin		Kaohsiung	Taiwan	22.67450	120.27950
Kaohsiung Mu	Sanmin		Kaohsiung	Taiwan	22.68030	120.27310
Sanmin Park	Sanmin		Kaohsiung	Taiwan	22.64600	120.30500
Sanmin Park	Sanmin		Kaohsiung	Taiwan	22.64600	120.30500
Kaohsiung mu	Gushan		Kaohsiung	Taiwan	22.65600	120.28600
Kaohsiung mu	Gushan		Kaohsiung	Taiwan	22.65600	120.28600
Aozihdi Forest	Gushan		Kaohsiung	Taiwan	22.65968	120.29839
Aozihdi Forest	Gushan		Kaohsiung	Taiwan	22.65968	120.29839
Sanmin DunSi	Sanmin		Kaohsiung	Taiwan	22.64919	120.29523
Sanmin DunSi	Sanmin		Kaohsiung	Taiwan	22.64919	120.29523
Day Lili Moun	Jingfeng		Taitung	Taiwan	22.63601	120.95336
Day Lili Moun	Jingfeng		Taitung	Taiwan	22.65139	120.95794
Uji Badland	Beinan		Taitung	Taiwan	22.81581	121.13829
Yang-Ming Pa	Taoyuan		Taoyuan	Taiwan	24.98210	121.30904
Yang-Ming Pa	Taoyuan		Taoyuan	Taiwan	24.98210	121.30904
Sha-Hsi Forest	Sandiman		Pingtung	Taiwan	22.76580	120.65800
Sha-Hsi Forest	Sandiman		Pingtung	Taiwan	22.76580	120.65800
Dashuichong	Sandiman		Pingtung	Taiwan	22.78630	120.64000
Dashuichong	Sandiman		Pingtung	Taiwan	22.78630	120.64000

ion Templa | CollectionMethodPickList

2.10. The values of these columns were set using a formula. You only need to choose the **Prep Type**

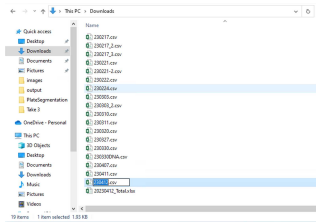
(Column W), which is the way you preserve your samples.

*Inform admin when other Prep Type is needed.



W	X	Y	Z
Prep Type	Prep Count	Room	Storage
Fridge	1	927	4C
Fridge	1	927	4C
Fridge	1	927	4C

2.15. Rename the file with the date you are going to upload the datasheet to Specify.



Upload your datasheet to Specify database

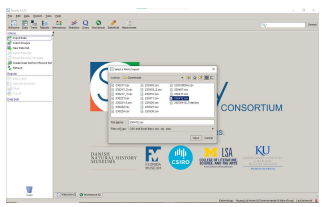
3.1. Go to the **Workbench** tab.



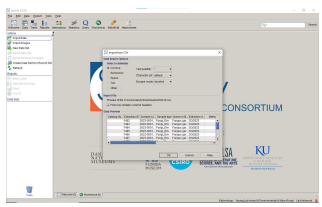
3.2. Click **Import Data**.



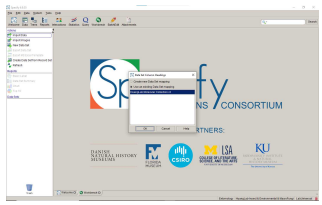
3.3. Choose the file you wanted to import (upload), and click **Open**.



3.4. Let everything as default. Then, click **OK**.

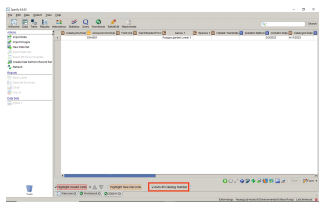


3.5. Select the second option: **Use an existing Data Set mapping**.

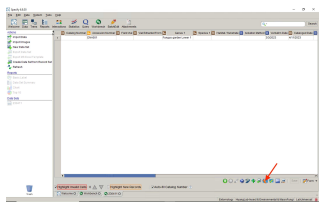


3.6. Check for any invalid cells (red highlighted cells).

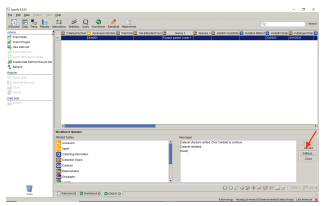
Remember to tick Auto-fill Catalog Number (marked with red box).



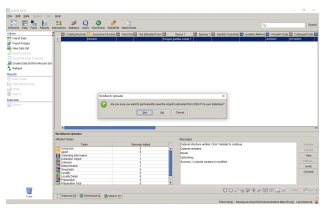
3.7. Click **Upload** (pointed with red arrow).



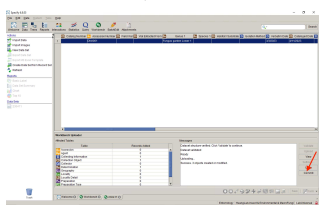
3.8. Click **Upload** again (pointed with red arrow).



3.10. Click **Yes** to finish the upload progress.

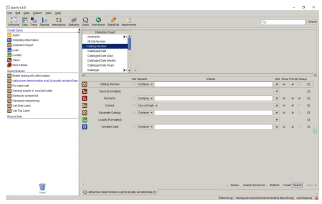


3.9. Then Click **Commit** (marked with red arrow).

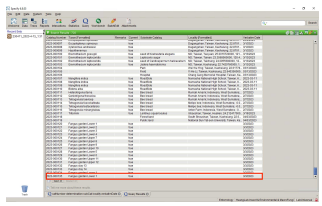


3.11. Now you can go to **Query** -> **catNumber determination…**(under the Saved Queries in left panel) to check if the upload succeeded.

Click **Search** to start querying.



3.12. Ta-Da!! The upload was succesful.

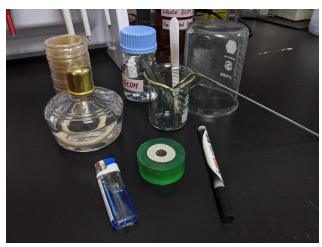


****Remember to delete the Data Sets & Record Sets before closing the Specify.**



Isolate the microbes from your environmental samples

4. Prepare the equipment and materials first for sample isolation. Put them on another bench first.



1. Alcohol lamps 2. 95% ethanol cooling bottle 3. Beakers (small & large) 4. Tweezers 5. Scalpel with blade 6. Separator 7. Grafting tube 8. Lighter 9. Marker pen 10. Media

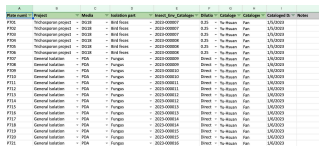
Upload your datasheet to Specify database

3.13. Paste the **Catalog Number** of uploaded data back to your spreadsheet.

Isolate the microbes from your environmental samples

4.3. Before isolation, key the information about your isolation (how many plates you are going to isolate, what kind of media you will use, etc) into the **1 Fungi Isolation Plates (Direct edit) 菌株分離培養基編號**

(直接編輯) spreadsheet first



Plate_num	Project	Media	Isolation part	Insect_Env_Cataloged #	Dilution	Cataloger First Name	Cataloger Last Name	Cataloged Date	Notes
P301	Trichosporon project	SN18	Bird feces	2023-000807	0.25	Yu-Xiuxin	Pan	1/1/2023	
P302	Trichosporon project	SN18	Bird feces	2023-000807	0.25	Yu-Xiuxin	Pan	1/1/2023	
P303	Trichosporon project	SN18	Bird feces	2023-000807	0.25	Yu-Xiuxin	Pan	1/1/2023	
P304	Trichosporon project	SN18	Bird feces	2023-000808	0.25	Yu-Xiuxin	Pan	1/1/2023	
P305	Trichosporon project	SN18	Bird feces	2023-000808	0.25	Yu-Xiuxin	Pan	1/1/2023	
P306	Trichosporon project	SN18	Bird feces	2023-000808	0.25	Yu-Xiuxin	Pan	1/1/2023	
P307	General Isolation	PDA	Fungus	2023-000809	Direct	Yu-Xiuxin	Pan	1/6/2023	
P308	General Isolation	PDA	Fungus	2023-000809	Direct	Yu-Xiuxin	Pan	1/6/2023	
P309	General Isolation	PDA	Fungus	2023-000810	Direct	Yu-Xiuxin	Pan	1/6/2023	
P310	General Isolation	PDA	Fungus	2023-000810	Direct	Yu-Xiuxin	Pan	1/6/2023	
P311	General Isolation	PDA	Fungus	2023-000811	Direct	Yu-Xiuxin	Pan	1/6/2023	
P312	General Isolation	PDA	Fungus	2023-000811	Direct	Yu-Xiuxin	Pan	1/6/2023	
P313	General Isolation	PDA	Fungus	2023-000812	Direct	Yu-Xiuxin	Pan	1/6/2023	
P314	General Isolation	PDA	Fungus	2023-000812	Direct	Yu-Xiuxin	Pan	1/6/2023	
P315	General Isolation	PDA	Fungus	2023-000813	Direct	Yu-Xiuxin	Pan	1/6/2023	
P316	General Isolation	PDA	Fungus	2023-000813	Direct	Yu-Xiuxin	Pan	1/6/2023	
P317	General Isolation	PDA	Fungus	2023-000814	Direct	Yu-Xiuxin	Pan	1/6/2023	
P318	General Isolation	PDA	Fungus	2023-000814	Direct	Yu-Xiuxin	Pan	1/6/2023	
P319	General Isolation	PDA	Fungus	2023-000815	Direct	Yu-Xiuxin	Pan	1/6/2023	
P320	General Isolation	PDA	Fungus	2023-000815	Direct	Yu-Xiuxin	Pan	1/6/2023	
P321	General Isolation	PDA	Fungus	2023-000816	Direct	Yu-Xiuxin	Pan	1/6/2023	

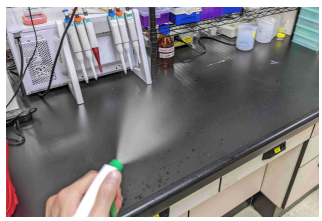
Plate number: The number is auto-generated. Project: Choose an existing one or create one for yourself for latter identified. Media: The media to isolate your microbes. Isolation part: The part of the environment samples you isolated your microbes. Insect_Env_Cataloged #: The catalog number of the environmental samples. Dilution: The dilution factor of this particular plate for isolation (e.g., 0.001, 0.01, or Direct). Cataloger First Name & Cataloger Last Name: The name of who uploads the data. Cataloged Date: The date on which the data is uploaded. Notes: Any information you wanted to remark on the certain plate. Leave it when you didn't have any notes.

4.1. Sterilize the microbes isolation area (air and bench) using a 75% ethanol sprayer.

****Make sure the mist is emitted.****



Spray it a few times up to the air.



Spray it a few times onto the bench.



Wipe the bench unidirectionally using paper towels.

4.2. Now, move the equipment and materials for microbes isolation back to the sterilized area.

4.4. Write the plate information on the downside of the plate following the example photo shown below.

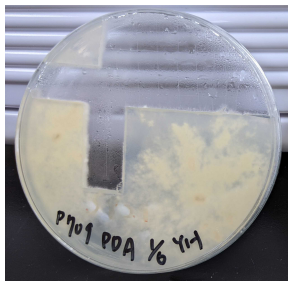


Plate number, Media, Date, Your name

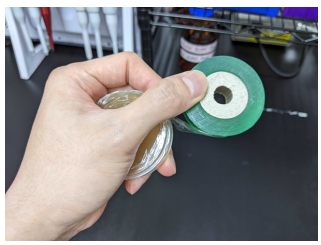
4.5. Make sure that you sterilize every tool before using them for each isolation:

1. Dip the scalpel, tweezer, or spreader into 95% ethanol cooling bottle, and flame it immediately.
2. Repeat the step(dip and flame) **THREE times** for sterilization of equipment.

4.6. Now, you can start to isolate microbes from your environmental samples. We called them primary cultures.

****Isolation method varied depend on the type of sample, check published papers or consult with seniors if needed****

4.7. After finishing all the isolation, seal the plates with grafting tape for 2 rounds.



4.9. Check the growth condition of the primary cultures every day. **Make sure they are not overgrown.** Once they are matured, you can start to subculture them.

4.8. Pack all the plates using a PP bag (no. 8, 10, or 12 depending on how many plates you have). Incubate the plates in your slot in the incubator for 7~21 days (depending on the growth rate of the microbes).

no. 8: 8 plates

no. 10: 24 or 32 plates

no. 12: 48 or 56 plates



Flip the PP bag over when keeping it in the incubator. The cover of the plates needed to be facing downwards to prevent the microbes from drowning by the water vapors.

Record and upload the data of primary cultures

5.

1. Examine the microbes colonies under a dissecting microscope.
2. Choose your desired colonies, which are your targeted colonies.
3. Mark the targeted colonies with their later corresponding secondary plate number.
4. If there is more than one morphotype targeted colony in a certain plate, you need to record them into a distinct plate number, e.g., **TWO morphotypes will have TWO distinct secondary plate numbers** from the same primary plate number.



5.1. Record all the data of primary cultures to the second worksheet: **Secondary Isolate PDA** (in 1 Fungi Isolation Plates (Direct edit) 菌株分離培養基編號 (直接編輯)).

Fill in all the columns following the photos and instructions shown below.

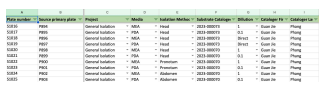
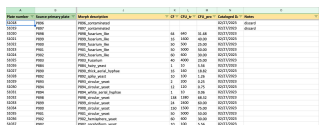


Plate number: It is auto-generated. Column B to Column I: You can copy and paste directly from the Column A to Column H of the first worksheet Primary Isolate MEA; (Hint: GREEN columns to GREEN columns)



Morph description: This is generated automatically by the formula. Replace YOUR DESCRIPTION with what you do observe on your targeted colony under a dissecting microscope. Type contaminated if you got others than your target or all the colonies were overgrown by common molds. CFU: The number of colonies of your target in a plate. CFU_true: It is auto-filled based on the dilution factor of each plate. CFU_proportion: The proportion of your targeted colony in a certain plate (i.e. what percentage of your target among all the observed colonies on

the plate). Cataloged Date: The date you do the subculture and upload the data. Notes: Any information you think is important to record. If you have contaminated in Morph description, please type discard here.

5.3. Before scanning plate, you need to sterilize the environment and scanner glass using paper towels with 75% ethanol.





5.5. Keep the order of the plates, remove the plates cover, and put the plates upside down onto the scanner glass.

****Do not change the order before and after scanning****

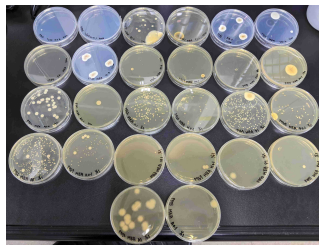


5.4. Wipe the scanner glass using paper towels with dishwashing detergent (this prevents water condensation on the glass).

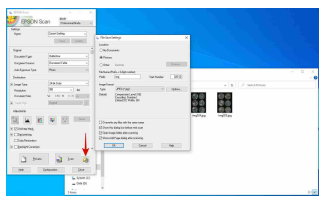




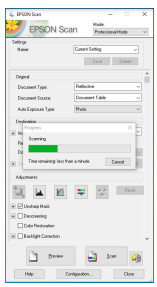
5.2. Record or take a photo of the order of your plates that are ready for scanning. **6** plates as a group, so you can refer back during the photo editing.



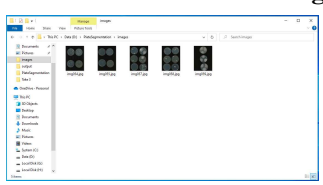
5.6. Search and open **EPSON Scan** software. Choose **Professional mode**. Go to the **File Save Settings** (red arrow), and make sure the image format is **.jpg** (follow all the settings as the photo below shown).



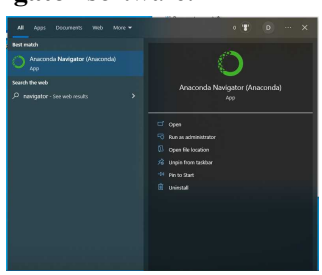
5.7. Then, click **OK**. (It will start scanning like the photo shown below).



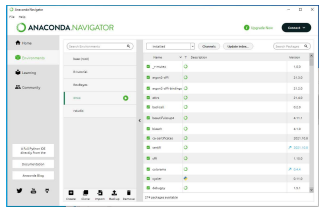
5.8. Move the files from your **Pictures** folder to the **D:\PlateSegmentation\images**



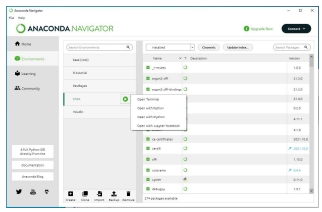
5.9. Search and open **Anaconda Navigator** software.



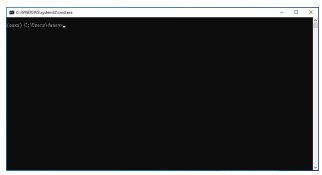
You will see the interface as the photo shown below.



5.10. Select **onxx** under **Environments** tab (as the photo shown below).

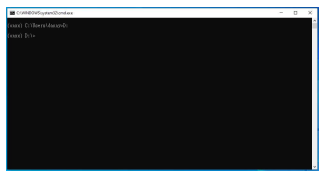


Left click the **play button**. Select **Open Terminal**.

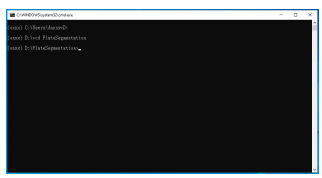


5.11. Follow the commands, and you will have the similar output as the photos shown below.

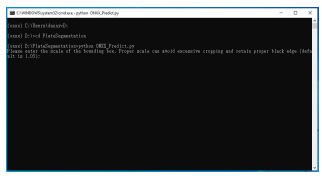
Type **D:** , then press **enter**



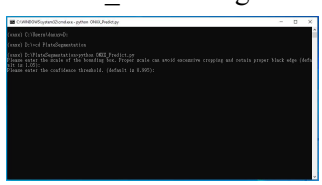
Type **cd PlateSegmentation** , then press **enter**

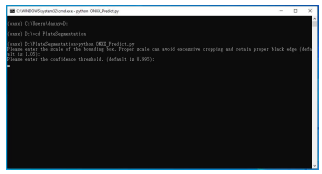


Type **python ONXX_predict.py** , then press **enter**

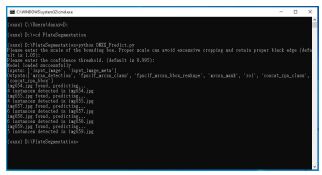


Press **enter** again and again. You will see a **_** flashing.

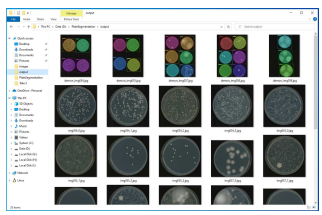




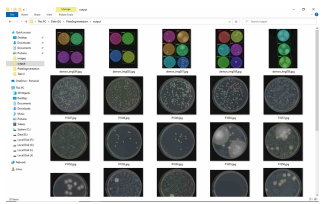
Then, a new command line comes out, which represent the program execution was complete.



5.12. The output files are in the **D:\PlateSegmentation\output**.



Rename the output files with their **corresponding plate number**.



Upload them to the **HuangLab Collection/Plate picture/20XX**



Subculture the primary cultures

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6. Follow (go to step 4.4) to (go to step 4.9) for the subculture of primary cultures. Now, we called them **secondary cultures**.

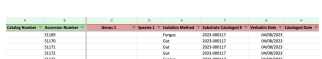
You may need to keep the primary cultures for a while before the secondary plate are all secured.

Ready for preservation

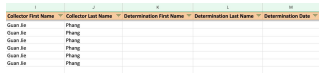
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7. After 7-21 days (depending on the growth rate of your targeted microbes), the secondary cultures are ready to be preserved. If the secondary cultures were contaminated, please redo the subculture from the primary cultures.

7.1. Before prepare preservation vials, the **For Specify cateloging COPY to MODIFY!!!** worksheet need to be filled.

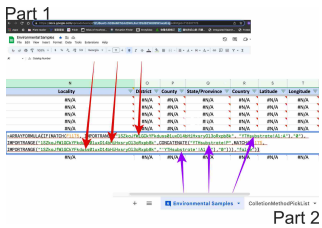


Catalog Number: It is auto-generated from Specify, fill in after you upload the data. Column B, Column E to Column G: It is synchronized from the Secondary Isolate PDA worksheet. Genus 1 & Species 1: Only fill in when you can identify them. Cataloged Date: Same as previous spreadsheet.



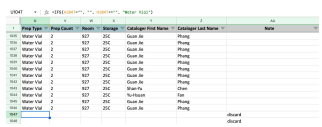
Collector First Name & Collector Last Name: Same as previous spreadsheet. Column K to Column M: Same as previous spreadsheet.

There is 2 part for all the locality information, you need to replace the each information: spreadsheet code (red arrow pointed), worksheet name (purple arrow pointed) with your own spreadsheet information.



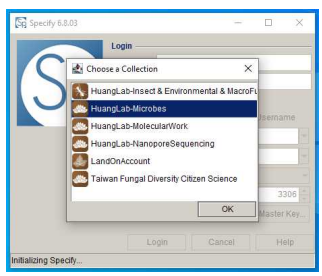
Part 1: Replace the red arrow pointed region of the written formula with your own spreadsheet code (highlighted region). Part 2: Replace the purple arrow pointed region of the written formula with your own worksheet name (blue shaded region).

If the Cataloged Date is filled, the Column U to Column Z is auto-filled with the written formula. Do not change it unless it needed. The Column AA is synchronized with the Column O of the **Secondary Isolate PDA** worksheet.



7.2. Once the spreadsheet is filled, download the data with .csv format.

Open **Specify**, go to **HuangLab-Microbes** collection.

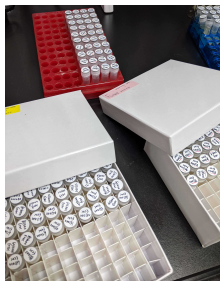


7.3. Follow (go to step 3.1) to (go to step 3.13) for preservation vials data upload.

7.4. For each **Catalog Number**, you must prepare **2** water vials and **2** glycerol vials. You also need a 2 ml microtube with lysis mixture and beads (for details, check **DNA extraction protocols**).

Each tube needed a sticker with its corresponding **Catalog Number** on top of the cap. The label should be written with a black color marker pen or a technical pen alternative.

Please follow the example in the photo shown below.



For water vials: either tube must have a RED underline. For glycerol vials: both tubes must have a BLUE underline.

The lysis mixture microtubes that are ready for use will look like this.



7.5. Operate UV sterilization of the laminar flow for 15 minutes. After UV sterilization, you need to ethanol-sterilize the materials that are going into the laminar flow.

Light up the alcohol lamp and flame the material.

****The materials you need are the same as (go to step 4) , but** you only need **a scalpel instead of a tweezer and a spreader.****

Culture preservation

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8. Preserve the plates with one water and one glycerol vial for one colony. The rest is for DNA extraction. Make sure you flame-sterilize the scalpel every time before slicing the agar.

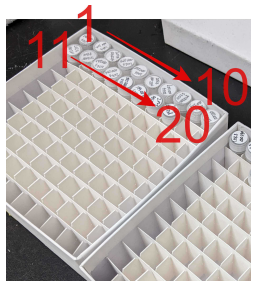
****The details of culture preservation are not provided here.****

8.1. Store the preservation vials in their corresponding freezing box.

For water vials, you can keep them directly.

For glycerol vials, you need to keep them in a freezing container and then freeze them inside a -20ºC fridge for 1 day before keeping them in the -80ºC fridge collection.

****The freezing boxes of water vials are kept in the steel cabinet in front of Prof. Huang's office; the freezing boxes of glycerol vials are kept in the -80ºC fridge outside N927****



The tubes are needed to be kept in the order shown above, but not the other way around.